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Nanocrystalline $Al_{87}Ni_{8.7}Y_{4.3}$ and $Al_{90}Fe_5Gd_5$ Alloys that Retain the Localized Corrosion Resistance of the Amorphous State

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The localized corrosion properties of $Al_{87}Ni_{8.7}Y_{4.3}$ and $Al_{90}Fe_5Gd_5$, were investigated in the amorphous, high strength partially nanocrystalline, and fully crystalline states. The critical pitting potentials and pit growth behaviors of amorphous and nanocrystalline alloys were improved compared to high purity, polycrystalline Al in 0.6 M NaCl solution. Transition metal (TM) and rare earth additions (RE) were retained in solid solution in amorphous alloys and the remaining amorphous matrix of partially nanocrystalline alloys. Substantial TM and RE solute rejection occurred from nano- and microcrystals. Notably, the improved resistance to pitting was retained in the partially nanocrystalline condition, but was completely lost in the fully crystalline alloys. The opportunity exists to create partially nanocrystalline Al-TM-RE alloys of great strength with corrosion resistance equal to their amorphous counterparts.

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It is well established in the literature that glassy alloys and alloys with a low degree of crystallinity offer superior resistance to corrosion and vastly improved mechanical properties over their conventional counterparts.[1,2] Amorphous alloys often have corrosion resistance one to two orders of magnitude greater than polycrystalline alloys with the same alloy composition.[1,2] One theory links the improved resistance of amorphous alloys to their ability to promote amorphous oxide formation. In high temperature gas, vitreous or amorphous oxides offer improved oxidation resistance due to the absence of oxide grain boundaries which provide a rapid diffusion path for concentration gradient driven ion movement.[3,4] In the case of metal passivity in aqueous solutions, ion transport is mostly driven by the electric field across the oxide film. The lack of oxide grain boundaries may lower ion migration rates, rendering the passive film more protective.[3-5] The rapid transport and accumulation of cation vacancies at the oxide/metal interface is one theory accounting for oxide breakdown.[6] Moreover, vitreous oxides on amorphous alloys perform well due to enhanced bond flexibility since the vitreous or amorphous material can rearrange to accommodate lattice mismatch and strain between the oxide and the metal.[5,7] As a result of this flexibility, almost all surface atoms can bond with oxygen or OH^- without requiring an optimal epitaxial relationship between an ordered metal substrate and oxide.[5] Intolerable changes in oxide/metal misfit strain with halide incorporation is another theory accounting for the rupture of protective oxide films on metals.[8] The improved corrosion resistance associated with the addition of 18% Cr to Fe is attributed, in part, to a change in the protective oxide structure from a well oriented spinel structure at 0-12% Cr to a noncrystalline structure at 18% Cr.[5] This disordering has been shown by low energy electron diffraction[9] and recently by scanning tunneling microscopy.[10] A retardation in ionic transport may occur because noncrystalline films have fewer defects or grain boundaries to enhance ionic movement. In summary, desirable amorphous oxide and amorphous alloy properties include defect minimization, film ductility, bond flexibility, and efficient, rapid film repassivation.[5,7]

In the case of aluminum alloys, the alloying with transition metals to supersaturated solid solution concentrations exceeding equilibrium solubilities has been a successful approach for improving localized corrosion resistance.[11-15] Here, the native oxide formed on crystalline Al is already amorphous and the main benefits of nonequilibrium alloying derive from two broad concepts. Alloying elements available in solid solution may be incorporated into the oxide film to enhance its resistance to halide chemisorption and/or penetration.[11,13] Alternatively, alloying elements in the underlying metallic substrate may reduce pit dissolution rates as a function of electrochemical potential,[16] increase the severity of critical depassivating pit solutions, or result in oxidized solute with limited solubility in the pit environment.[17]

Ion implantation, codeposition by evaporation or sputtering, and rapid solidification are methods to obtain both nonequilibrium alloying and an amorphous condition. High strength can be obtained in the amorphous state without precipitation age hardening in the case of the class of Al-transition metal-rare earth (Al-TM-RE) alloys.[18-25] The intermetallic phases in Al-based precipitation age hardened alloys typically severely degrade passivity and corrosion resistance due to local galvanic coupling between precipitates and the matrix.[26] Strength is raised not only when the alloying elements in Al-TM-RE alloys are retained in a disordered solid solution and precipitate formation is suppressed, but is particularly improved upon formation of small nanocrystals in an amorphous matrix.[27-29] However, partial or complete crystallization often significantly reduces the corrosion resistance of amorphous alloys.[30,31] Therefore, the goal of this preliminary study was to ascertain whether Al-TM-RE glasses with nanocrystalline embedded in an amorphous matrix could retain the improved corrosion resistance of their amorphous counterparts.

Experimental

Rapid solidification by deposition of molten alloys on a cooled wheel was used to produce amorphous metal ribbons.[24,32] Alloy compositions of $Al_{87}Ni_{8.7}Y_{4.3}$ and $Al_{90}Fe_5Gd_5$ were selected based on prior evidence of amorphous behavior.[24,32] Preliminary studies over a range of TM and RE compositions with greater than 80 atom % Al produced negligible differences in localized corrosion behavior. Selected area diffraction, as well as transmission electron microscopy, confirmed a completely amorphous state in the as-solidified condition for both alloys (Fig. 1a and b). Ribbons of cross-sectional dimensions 0.15 by 0.002 cm were stored at 0°C until testing. Partially nanocrystalline, amorphous matrix variants were produced by an isothermal heat-treatment in evacuated quartz tubes at 150°C for 25 h (Fig. 2a and b).[33] Nanocrystals of approximately 10-15 nm diam were embedded in an amorphous matrix for the $Al_{90}Fe_5Gd_5$ alloy. Nanocrystals possessed a composition >98 atom % Al and a face-centered cubic (fcc) crystal structure. The $Al_{87}Ni_{8.7}Y_{4.3}$ alloy had a slightly broader range in Al nanocrystal diameters (10-25 nm). Transition and rare earth metals were rejected from nanocrystals in both alloys owing to their poor solubility in fcc Al.[33] The Fe or Ni concentrations reached levels in the remaining amorphous matrix of the partially nanocrystalline state that were slightly above completely amorphous levels due to solute rejection from nanocrystals. This concentration decreased sharply at the nanocrystal interface. The rare earth metal (Y, Gd) concentrations reached maxima at the interface of approximately 9 atom % owing to slow solid-state diffusion

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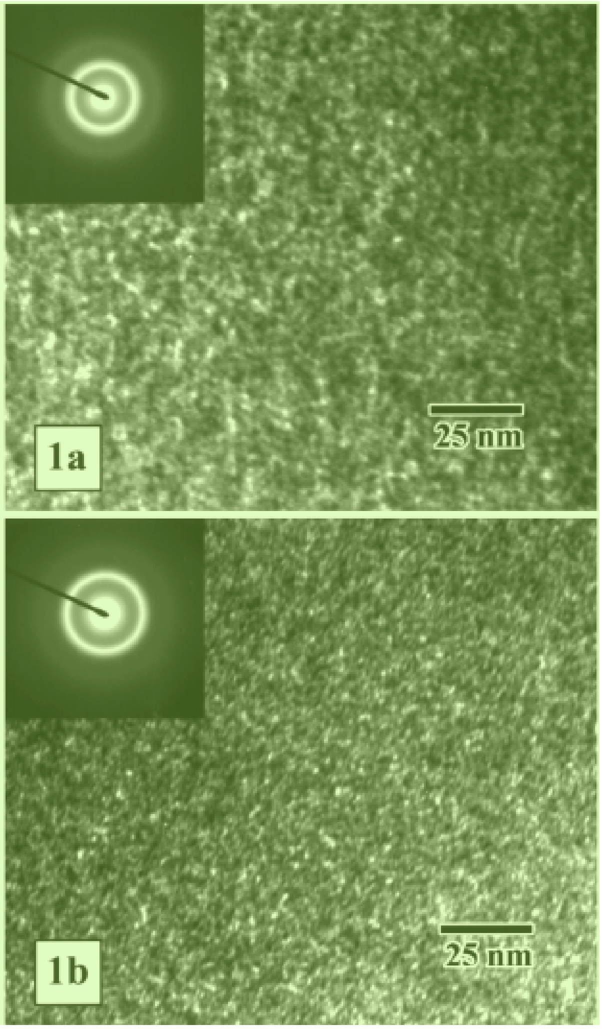

Figure 1. Darkfield transmission electron micrographs with diffraction pattern (inset) for (a) $Al_{87}Ni_{8.7}Y_{4.3}$ and (b) $Al_{90}Fe_5Gd_5$ in the as-formed fully amorphous state.

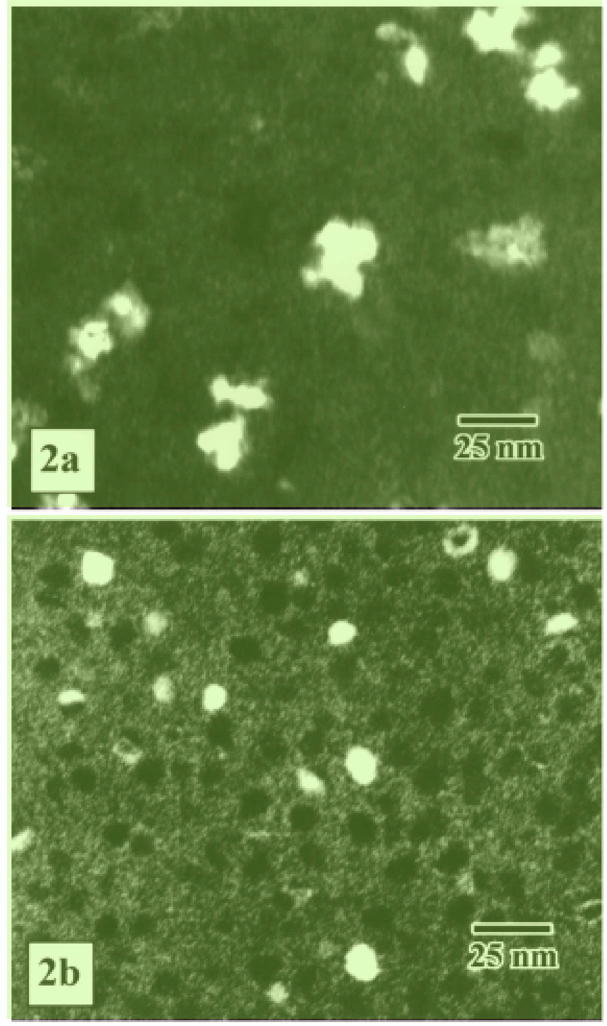

Figure 2. Dark-field transmission electron microscope images of partially nanocrystalline-amorphous matrix (a) $Al_{87}Ni_{8.7}Y_{4.3}$ and (b) $Al_{90}Fe_5Gd_5$ illustrating the Al-rich nanocrystal distribution in the amorphous Al-TM-RE matrix.

of Gd and Y.[33] Y and Gd concentrations then decreased within the amorphous matrix to levels just above those of the wholly amorphous condition. Embedding 10 nm diam nanocrystals in an amorphous matrix produces high tensile fracture strength, high specific strength compared to conventional precipitation age hardened Al-base alloys (Table I), and good ductility.[27-29] Fully microcrystalline alloys were created by heat-treatment at 550°C for 1 h (Fig. 3a and b).[33] Al-rich fcc microcrystals were almost completely depleted of TM and RE collected in Al-TM and Al-RE intermetallic phases. Polycrystalline, high purity (99.999% Al) foil with an as-drawn grain diam of 10-100 μm was utilized as a control.

A conventional three-electrode configuration was utilized to determine pitting (E_p) and repassivation (E_r) potentials in nitrogen purged 0.6 M NaCl solution (pH 5.9). A saturated calomel reference electrode (SCE) was utilized and experiments were conducted at 25°C. Electrode surface areas of 10^{-4} cm^2 were utilized by mounting melt-spun ribbon cross sections in Buehler epoxide resin and polishing to a 0.05 μm finish. High purity Al foil of similar dimensions was tested to eliminate any possible variations in E_p with electrode size owing to changes in the exposed, at-risk surface area.[34] The typical experimental procedure involved a 1 h exposure at open-circuit potential, an anodic scan at 0.1 mV/s, and by a reverse in scan direction at a 0.05 mA. E_p potentials were determined from achievement of a current density (cd) two orders of magnitude above the passive cd during the anodic scan. Repassivation potentials were obtained during the reverse scan at the potential where an abrupt decrease in

Table I. Typical tensile strength of fully amorphous and partially nanocrystalline-amorphous matrix $Al_{90}Fe_5Gd_5$ and $Al_{87}Ni_{8.7}Y_{4.3}$ alloys compared to a conventional Al-base precipitation age hardened alloy (from Ref. 24, 27, and this study).

Material	Microstructure	Tensile strength (MPa)	Density (g/cm^3)
$Al_{90}Fe_5Gd_5$	Amorphous[a]	735	3.2
$Al_{90}Fe_5Gd_5$	Nanocrystals in amorphous matrix[a,b]	1010	-
$Al_{87}Ni_{8.7}Y_{4.3}$	Amorphous[a]	880	3.3
AA 2024-T3	Crystalline[a]	485	2.8

[a] All alloys exhibited excellent ductility in 180° bend tests.
[b] Nanometer diamater fcc crystals embedded in an amorphous matrix.

anodic current density to well below 0.5 mA/cm^2 was achieved. Pit growth kinetics were studied using initially flush mounted electrodes (10^{-4} cm^2) that were electrochemically dissolved to depths of 0.5-0.7 mm by initial anodic polarization to +1 V_{SCE} for 1000 s. Such artificial pits were, subsequently, immediately stepped down to applied potentials near the pitting potential of anodic scans and subjected to further sequence of potential steps and holds: 50 mV for 25 s at potentials above the value of E_p determined from anodic scans,

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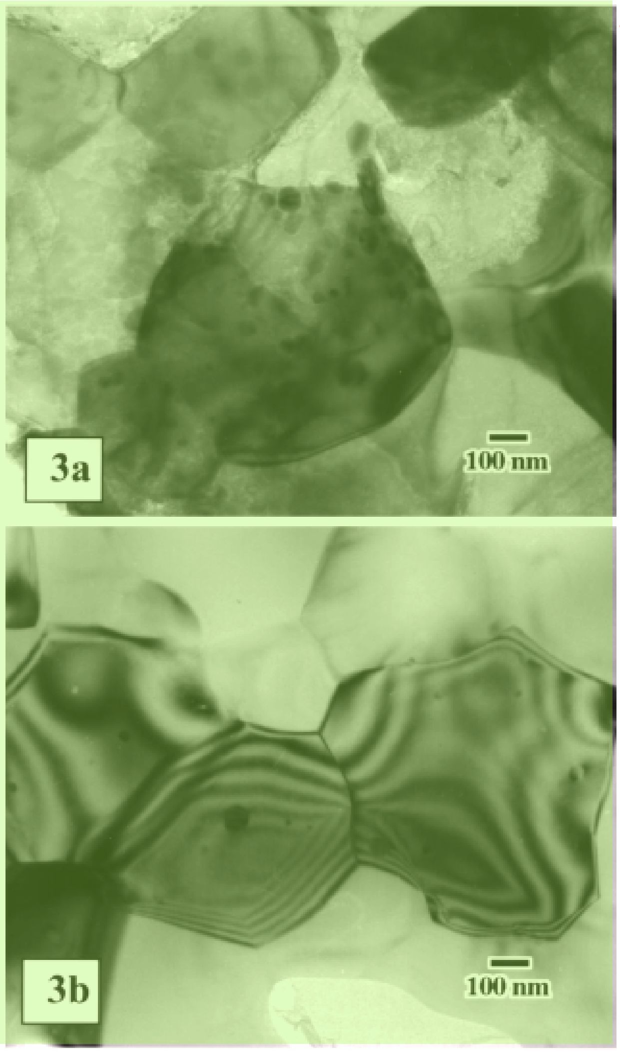

Figure 3. Transmission electron micrograph of the fully microcrystalline (a)$Al_{87}Ni_{8.7}Y_{4.3}$ (dark field) and (b) $Al_{90}Fe_5Gd_5$ (bright field). $Al_{90}Fe_5Gd_5$ results are from Ref. 33.

25 mV for 40 s at intermediate potentials, and 10 mV for 25 s until applied potentials were lowered below E_r from anodic scans. The average current was constant over time during these holds but decreased with potential. The short times during each potential step and, thus, small anodic charge served to limit additional change in pit depths after the initial depth established at +1 V_{SCE}. This enabled construction of E- log(i) curves at nearly constant pit depth. The critical potential and current density associated with repassivation of artificial pits was recorded as an abrupt decrease in anodic current and current noise associated with cessation of hydrogen evolution. *E-i* data from artificial pits were corrected for electrolyte solution resistance using high frequency impedance measurements. *E-i* data were also corrected for cathodic current in pits that was not measured by the potentiostat. Anodic current efficiency factors were determined by comparing actual (by post-test depth measurement) and electrochemically estimated (by integration of net current, i_{net}, to yield net charge) artificial pit penetration depths. The current efficiency factors (i_{net}/i_{anodic}) were 0.83 for pure Al, 0.66-0.71 for amorphous alloys, 0.63-0.64 for nanocrystalline alloys, and 0.50-0.54 for the crystalline alloys. The alloy equivalent weights and densities used for electrochemical depth calculations were 9.67 g/equiv and 3.2 g/cm^3 for $Al_{90}Fe_5Gd_5$, 9.89 g/equiv and 3.3 g/cm^3 for $Al_{87}Ni_{8.7}Y_{4.3}$, and 8.99 g/equiv and 2.7 g/cm^3 for high purity Al. Pit bottom areas were smooth enough to enable straightforward calculation of current density. The resulting data are reported as E_{true} vs. log i_{anodic} where applied potential is ohmically corrected using i_{net}, and i_{anodic} accounts for cathodic reactions in the pit through use of current efficiency factors.

Results

E_p and E_r were elevated on amorphous and partially nanocrystalline $Al_{90}Fe_5Gd_5$ relative to polycrystalline Al as shown in Table II. E_r was raised on amorphous and partially nanocrystalline $Al_{87}Ni_{8.7}Y_{4.3}$. E_r was much more reproducible than E_p. Note that E_p values were also elevated on 10^{-4} cm^2 surface area Al electrodes compared to larger surface area Al electrodes owing to the smaller at-risk surface area available for pitting.[34] E_p values obtained for larger Al electrodes were in excellent agreement with the literature.[26] E_p was elevated on amorphous and partially nanocrystalline $Al_{90}Fe_5Gd_5$ even after accounting for electrode surface area effects by conducting experiments on Al electrodes of the same surface area. In contrast, E_r was independent of initial electrode surface area and pits of similar depths were repassivated in most cases. This is important because E_r often depends on pit depth[35] and variations in pit depth from test to test would thwart attempts to make careful material comparisons. Clearly, the presence of TM and RE in the amorphous alloy matrix results in enhanced E_r relative to Al. This is seen both in the amorphous and partially nanocrystalline conditions for both alloys (Table II).

Artificial pit studies (Fig. 4a and b) indicated that significantly more positive true anodic potentials were required to achieve the same pit dissolution rates for Al in amorphous and partially nanocrystalline alloys. Moreover, the anodic current density at which pits repassivated, i_{crit}, was elevated in amorphous and partially nanocrystalline alloys even though all pits were of similar depths. In contrast, fully microcrystalline alloys behaved similarly to polycrystalline Al. The observed differences in E_{true} vs. log i_{anodic} may reflect intrinsic alterations in material pit growth properties at nearly constant artificial pit depth.

Discussion

E_p, E_r, and artificial pit growth studies all indicate improvement in resistance to localized corrosion initiation and/or stabilization, slower pit growth, as well as easier repassivation in amorphous alloys. Figures 4a and b indicate that E_{true} vs. log i_{anodic} data are shifted to substantially more positive potentials and i_{crit} is greater in the amorphous condition. Such improvements may be attributed to either TM and RE alloying elements in solid solution and/or amorphous structure. Note that pit growth in partially nanocrystalline artificial pits is suppressed nearly as much as in fully amorphous artificial pits (i.e., it exhibits a similarly enhanced E_r, requires high true potentials to produce a given pit growth rate, and has high values of i_{crit}). TM and RE solute are depleted from the grains of fully microcrystalline alloys and collected at intermetallic phases. Here, active pit dissolution kinetics predictably approach those of polycrystalline Al and improved resistance to pit dissolution is completely lost.

In the context of critical chemistry models for pit stabilization,[36] E_r data suggest but does not prove that either the pit growth potentials required to maintain depassivation and/or the severity of the critical depassivating pit solution may be altered by amorphous state and/or retention of alloying elements in solid solution. The artificial pit growth data (Fig. 4a and b) suggest that different E_{true}-log(i_{anodic}) pit growth behavior is involved in improving repassivation properties for the amorphous and partially nanocrystalline alloys. Achievement of the same pit dissolution rates as seen for crystalline Al requires 400 mV more positive pit growth potentials. A similar effect was observed for crystalline Fe-Cr-Ni artificial pits when alloyed Mo content was increased[37] and Al-TR alloys.[16] In this study, cessation of active pitting occurs at a higher i_{crit}, suggesting some influence of TM and RE in solid solution. According to Galvele's criterion,[36] repassivation occurring at a higher i_{crit} for a similar pit depth suggests that repassivation occurs in a more concentrated pit solution.

Regarding the partially nanocrystalline alloys with nanocrystals embedded in an amorphous matrix, it is reasonable to assert that pit

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Table II. Pitting (E_p) and repassivation (E_r) potentials for $Al_{90}Fe_5Gd_5$ and $Al_{87}Ni_{8.7}Y_{4.3}$ at the 99% confidence level. Potentials reported are in volts vs. a saturated calomel electrode (V_{SCE}) at 25°C.

Parameter	Pure Al[a]	Amorphous $Al_{90}Fe_5Gd_5$ [$Al_{87}Ni_{8.7}Y_{4.3}$]	Nanocrystal $Al_{90}Fe_5Gd_5$ [$Al_{87}Ni_{8.7}Y_{4.3}$]	Fully Crystalline $Al_{90}Fe_5Gd_5$ [$Al_{87}Ni_{8.7}Y_{4.3}$]
E_p (V vs. SCE)	-0.376 ± 0.113	-0.099 ± 0.093	+0.044 ± 0.130	-0.499 ± 0.163
E_r (V vs. SCE)	-0.754 ± 0.009	-0.304 ± 0.007 [-0.358 ± 0.011]	-0.334 ± 0.009 [-0.371 ± 0.018]	-0.798 ± 0.006 [-0.760 ± 0.008]

[a] 10^{-1} cm^2 Al electrodes produced E_p and E_r of -0.649 ± 0.034 and -0.805 ± 0.033 V, respectively.

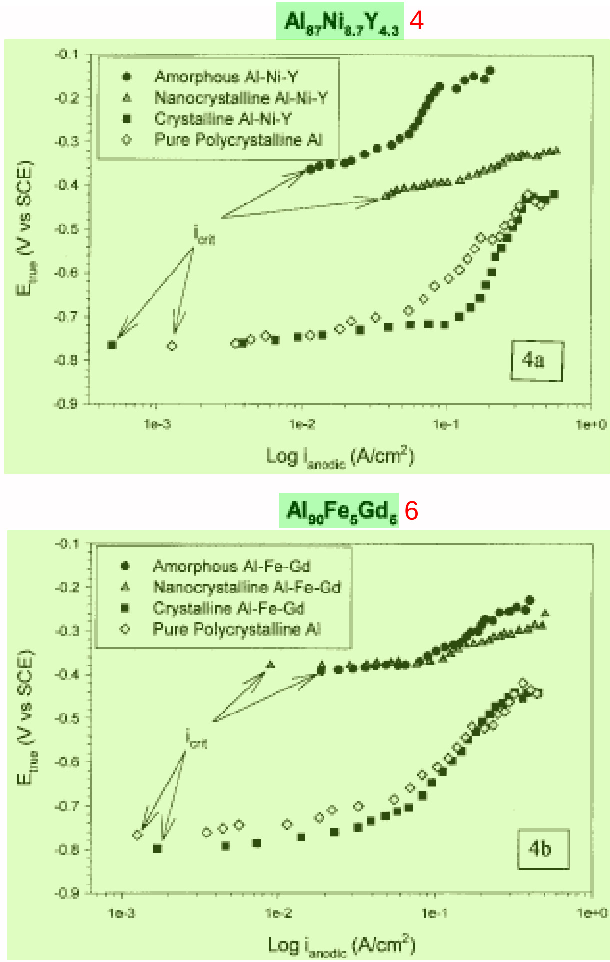

Figure 4. Ohmically corrected pit growth potential vs. anodic current density for (a) $Al_{87}Ni_{8.7}Y_{4.3}$ and (b) $Al_{90}Fe_5Gd_5$ artificial pits compared to pure Al. (● amorphous alloy), (◇ pure Al), (▲ partially nanocrystalline alloy), (■ crystalline alloy). The i_{crit} indicates the active pit cd just before repassivation.

initiation will occur more readily at Al-rich nanocrystals that lack TM or RE solute. However, such pits would, very quickly, experience hindered growth due to the solute gradient that exists at the nanocrystal/amorphous interface, assuming that the behavior in Fig. 4 pertains to this situation. TM and RE in solid solution create the need for even higher anodic potentials to achieve the same rate of Al^{+3} cation production as in Al. In critical chemistry models of pitting Al^{+3} production must be high enough to enable sufficient hydrolysis and Cl^- migration to maintain a depassivating pit solution despite the highly destabilizing concentration gradient in the liquid solution at pits.[36] Therefore, nanopits that grow to the nanocrystal/amorphous interface would require much higher growth potentials than Al to sustain active pit growth. Direct evidence for nanopits at nanocrystals embedded in an amorphous matrix is being sought.

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